

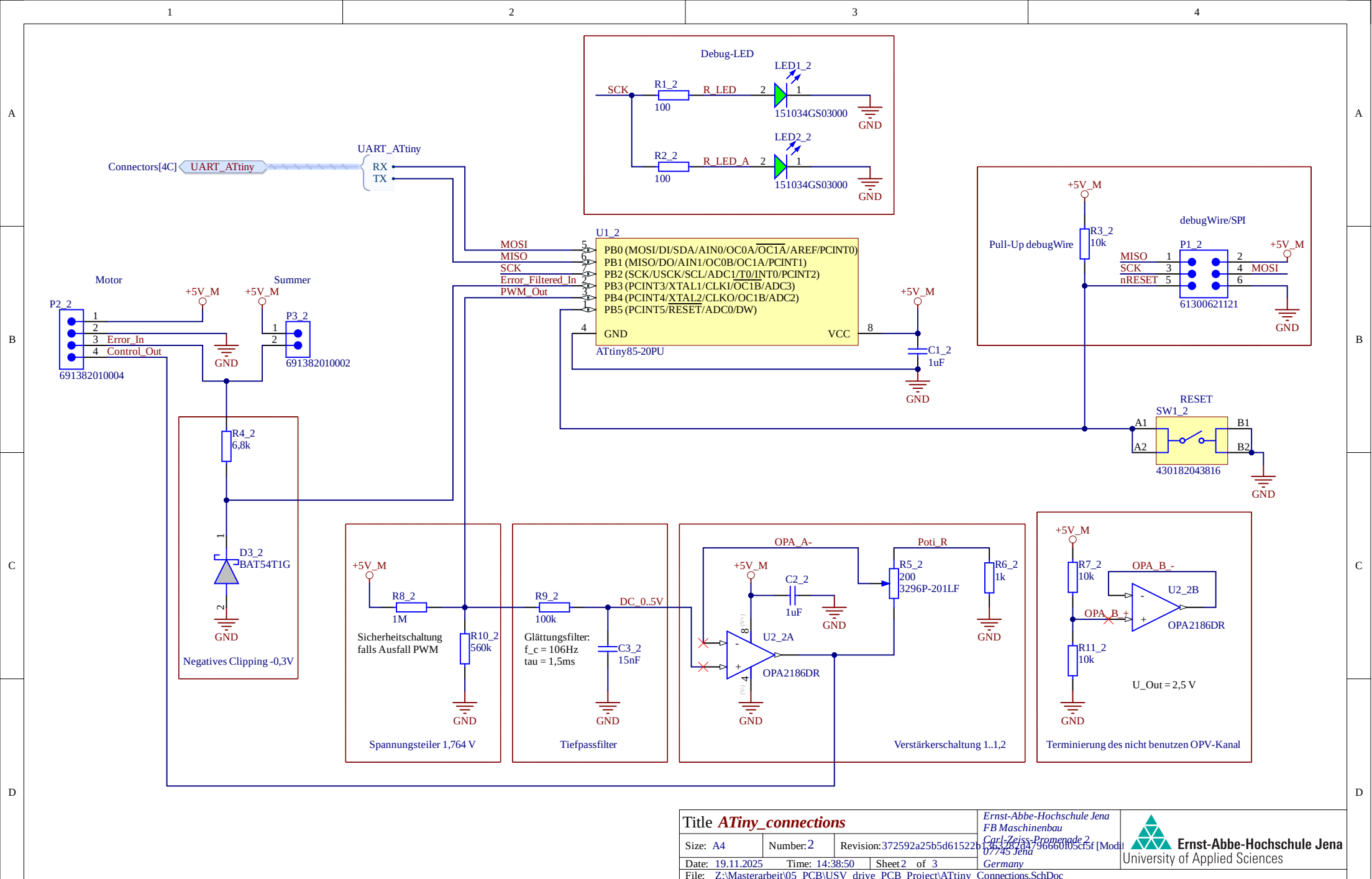
Overview for USV_drive

Sheet 1: ATtiny connections

Sheet 2: Raspberry Pi Pico H (RP2040) and Bluetooth-Modul HC-06 connections

Title Overview_USV_drive			Ernst-Abbe-Hochschule Jena FB Maschinenbau
Size: A4	Number: 1	Revision: 8d99f485fe26f059ea7e9d3a5236aa0504d7 [Modified: 19.11.2025 14:38:50]	Carl-Zeiss-Promenade 2 07745 Jena Germany
Date: 19.11.2025	Time: 14:38:50	Sheet 1 of 3	
File: Z:\Masterarbeit\05_PCB\USV_drive_PCB_Project\Overview.SchDoc			





A

A

B

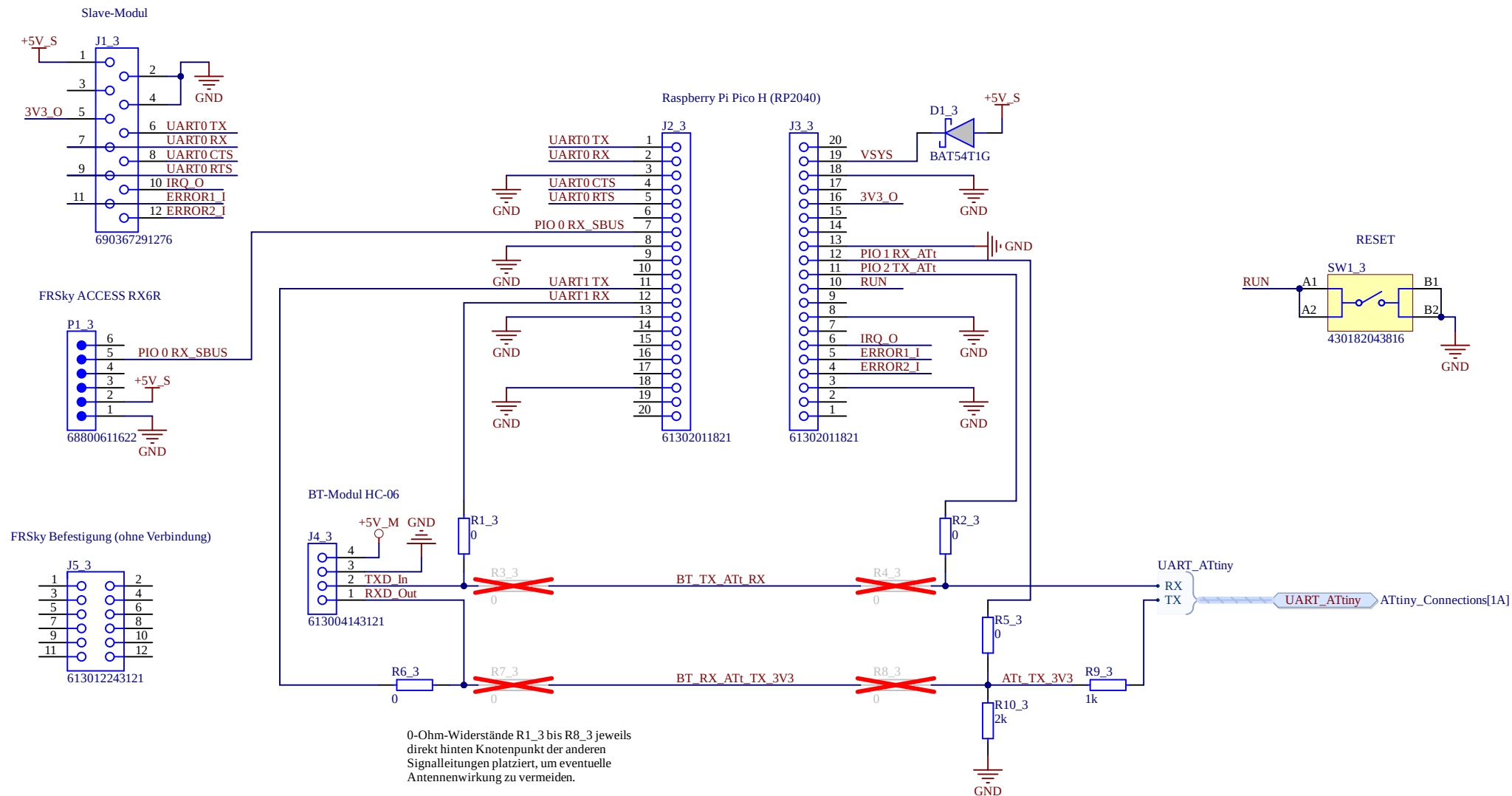
B

C

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D

D

Title **RP2040 Pico and HC-06 connections**

Ernst-Abbe-Hochschule Jena
FB Maschinenbau
Carl-Zeiss-Promenade 2
07745 Jena
Germany



Ernst-Abbe-Hochschule Jena
University of Applied Sciences

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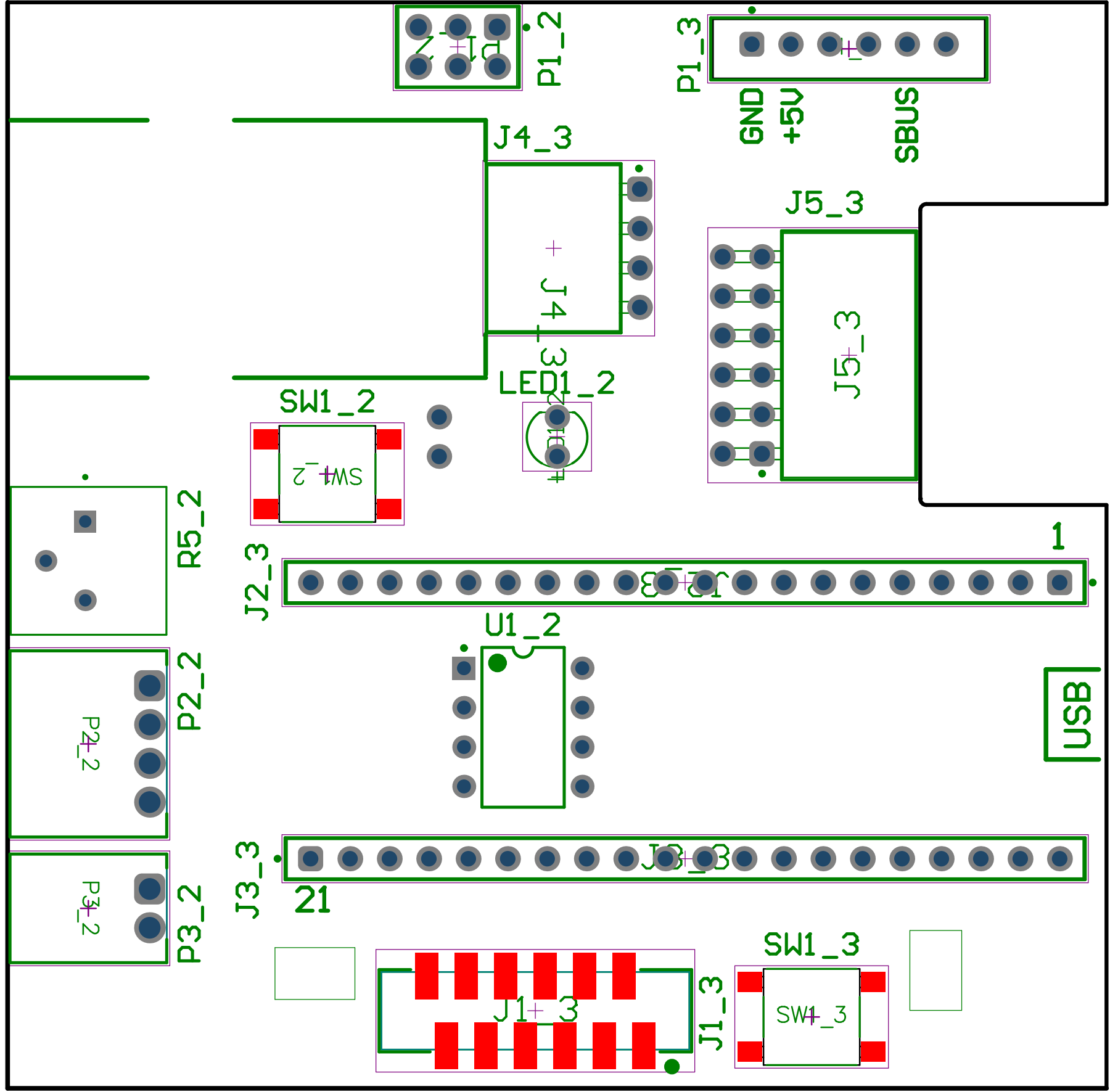
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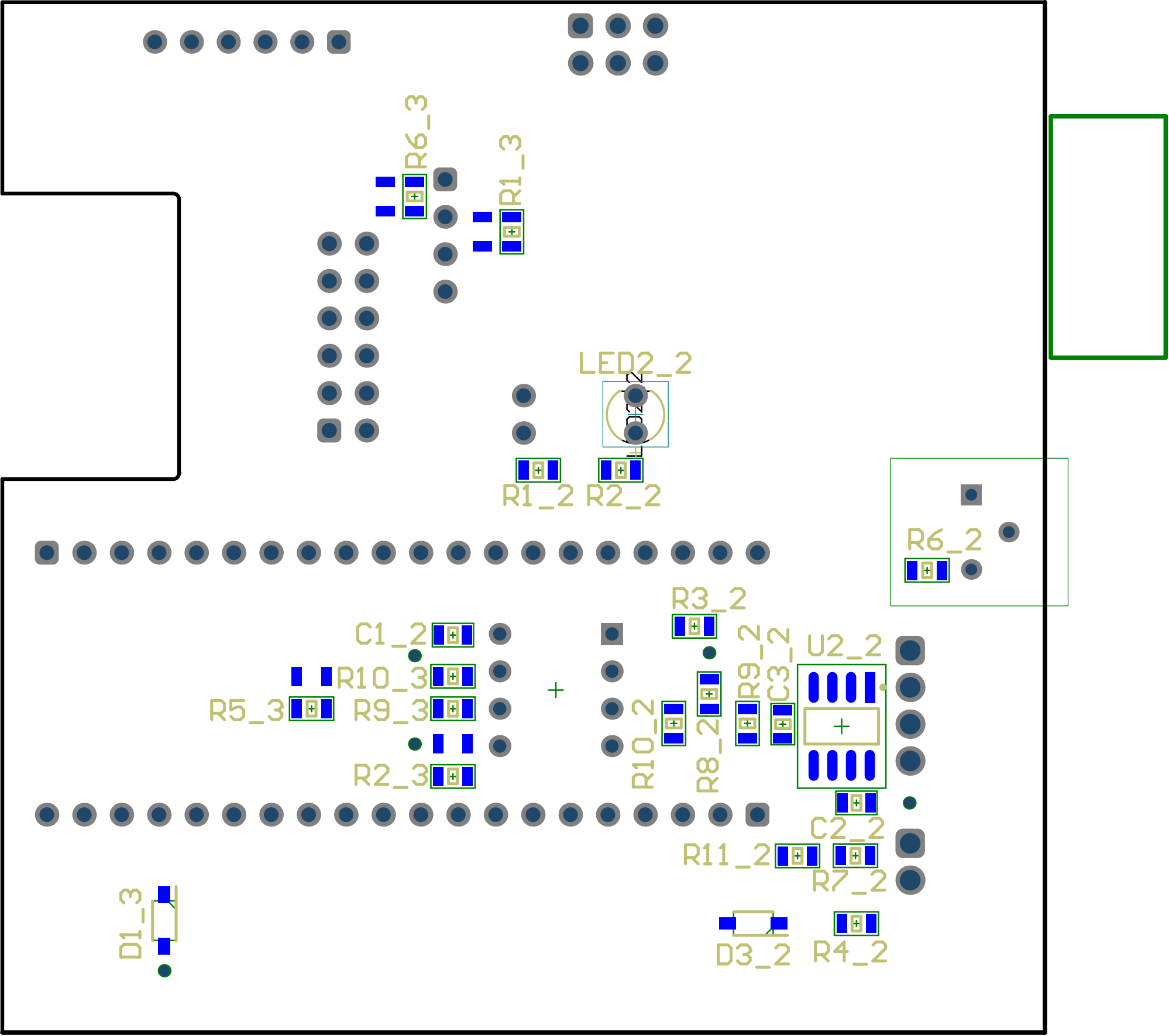
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Sheet 3 of 3

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Name	Description	Designator	Quantity	Manufacturer Part Number	Manufacturer Part Number 1
1uF	Kondensator	C1_2, C2_2	2		
15nF	Kondensator	C3_2	1		
BAT54T1G	S Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q101 Qualified and PPAP Capable; Low Forward Voltage 350 mV (Typ) @ IF = 10 mA; Extremely Fast Switching Speed; These Devices are Pb-Free, Halogen Free/BFR Free and are RoHS Compliant	D1_3, D3_2	2		
690367291276	WR-MM Mini Module Connector, Female, SMT, with Latch & Polarization, Straight, pitch 2.54mm, 12p	J1_3	1	690367291276	
61302011821	WR-PHD Socket Header, THT, Vertical, pitch 2.54mm, 1 Row, 20P	J2_3, J3_3	2	61302011821	
613004143121	WR-PHD Socket Header, THT, Angled, pitch 2.54mm, 1 Row, 4P	J4_3	1	613004143121	
613012243121	WR-PHD Socket Header, THT, Angled, pitch 2.54mm, 2 Row, 12P	J5_3	1	613012243121	
Green, 3.2V, 30°	WL-TMRW Mono-color Round Waterclear, THT, D3.9 x H5.3mm, Green, 3.2V, 30°	LED1_2, LED2_2	2	151034GS03000	
61300621121	WR-PHD Pin Header, THT, Vertical, pitch 2.54mm, 2 Row, 6P	P1_2	1	61300621121	
68800611622	WR-WTB "J" Type THT Vertical Male Shrouded Header, pitch 2.5 mm, 6 pins	P1_3	1	68800611622	
691382010004	WR-TBL Series 382 Horizontal PCB Header, THT, pitch 2.5mm, 04p	P2_2	1	691382010004	
691382010002	WR-TBL Series 382 Horizontal PCB Header, THT, pitch 2.5mm, 02p	P3_2	1	691382010002	
100	Widerstand	R1_2, R2_2	2		
0	Widerstand	R1_3, R2_3, R5_3, R6_3	4		
10k	Widerstand	R3_2, R7_2, R11_2	3		
6,8k	Widerstand	R4_2	1		
200	Trimmer Potentiometers Lead Sealed Type Multiturn Trimming Potentiometer Resistance range: 200 Ohm 25 Turns	R5_2	1		3296P-1-201LF
1k	Widerstand	R6_2, R9_3	2		
1M	Widerstand	R8_2	1		
100k	Widerstand	R9_2	1		
560k	Widerstand	R10_2	1		
2k	Widerstand	R10_3	1		
430182043816	WS-TASV SMT Tact Switch, 6mm x 6mm, 160gf, SPST-NO, 12V, 50mA	SW1_2, SW1_3	2	430182043816	
ATtiny85-20PU	8-bit AVR Microcontroller, 20MHz, 2.7-5.5V, 8K Flash, 512 Bytes EEPROM, 512 Bytes SRAM, Industrial Grade (-40°C to 85°C), 8-Lead PDIP, Pb-Free	U1_2	1		

Name	Description	Designator	Quantity	Manufacturer Part Number	Manufacturer Part Number 1
OPA2186DR	The OPA186, OPA2186, and OPA4186 (OPAx186) are low-power, 24-V, rail-to-rail input and output, zerodrift operational amplifiers (op amps). These op amps feature only 10 μ V of offset voltage (maximum) and 0.04 μ V/ $^{\circ}$ C of offset voltage drift over temperature (maximum).	U2_2	1		OPA2186DR